

Course “Object-Oriented Programming in C++”**Week №12*****Homework*****Exercise1.**

Create a base class "Pet" and derived classes "Dog", "Cat", "Parrot". Set the name of each animal and its characteristics using a constructor.

Exercise2.

Describe the base class "String" using the following methods:

- 1) Constructor without parameters;
- 2) Constructor that takes as a parameter C- string (it ends with a null byte);
- 3) Copy constructor;
- 4) Assignment operator;
- 5) Get a string length;
- 6) Clean up a string (to make a string empty);
- 7) Destructor;
- 8) Strings concatenation (overloading operators + and +=);
- 9) Check for equality (=) and inequality (!=).

Describe the class "Bit string" that is derived from the class "String"

(Strings of this class can contain only characters '0' and '1'). If in a basis of the initializing strings any characters will appear, that differ from allowable, the "Bit string" becomes empty. The content of the string is treated as a binary representation

of an integer with a sign bit. Negative numbers are stored in an additional code.

Methods:

- 1) Constructor without parameters;
- 2) Constructor that takes as a parameter C- string;
- 3) Copy constructor;
- 4) Assignment operator;
- 5) Destructor;
- 6) Change a sign of a number (to transfer a number to the additional code);
- 7) Addition of bit strings (overloading operators + and + =);
- 8) Check for equality (= =) and inequality (!=).

Exercise 3.

Using a template inheritance mechanism, create a hierarchy of classes. Base class called base has the following member variables:

1. T1 value1; // certain value
2. T2 value2; // certain value.

A descendant named child has additional member variables:

1. T3 value3;
2. T4 value4.

A descendant named child2 adds such member variables:

1. T5 value5;
2. T6 value6.

Each class has a required set of constructors, destructors, member functions to work with a class structure.